NephroTwin

A Personalized Digital Twin Prototype for Heat-Associated Physiological Strain in Early-Stage CKD

- Team: REPLACE
- College/incubator: REPLACE
- Challenge: Happiest Health Digital Twin Challenge 2026

Problem & motivation

Heat exposure, activity and baseline health may interact differently across individuals with CKD.

- Aim: explore personalized physiological-strain trajectories
- Prototype target: illustrative index over a 24-hour hypothetical horizon
- Not a diagnosis or prediction of confirmed kidney injury

Proposed solution

A virtual patient that combines historical profile information with changing time-series inputs.

- Historical profile: age, fictional CKD label, baseline eGFR and history
- Dynamic stream: heart rate, steps, temperature, humidity, simulated hydration
- Dashboard: observed trends and hypothetical scenario comparison

Architecture

Synthetic data → ingestion → virtual state/index → scenario simulator → dashboard.

- Python, pandas, NumPy
- Streamlit and Plotly
- Tests and reproducible generation

Data design

All included data are synthetic and generated with a fixed random seed.

- One fictional profile
- 168 hourly sensor/environment records
- No real patient data; no claim of physiological or clinical validity

Model design

A transparent, hand-set weighted index demonstrates the end-to-end workflow.

- Bounded 0–100 illustrative index
- 24-hour hypothetical scenario simulation
- Changing scenario inputs changes the trajectory
- Weights are not medically validated

Dashboard demonstration

Demonstrate profile, sensor trends, index over time, and scenario controls.

- Show baseline scenario
- Increase hypothetical temperature or activity
- Compare the simulated trajectories
- Explain what the score does and does not mean

Evaluation & limitations

Current prototype validates software flow, not clinical performance.

- Basic software tests check output bounds and 24-hour forecast length
- Synthetic labels cannot establish medical accuracy
- No real clinical outcome prediction or calibration

Responsible use & future work

The prototype is for research demonstration only.

- Obtain appropriate labelled data and clinical oversight
- Use patient-level splits and external validation
- Assess calibration, uncertainty, and subgroup performance
- Avoid medical advice or automated treatment decisions

Conclusion

NephroTwin demonstrates how a virtual patient can combine static history and dynamic signals for scenario exploration.

- Repository: add public GitHub URL
- Demo video: add accessible link
- Team: replace placeholders before submission